Linear Response formulation

$$H_{int} = \frac{1}{2\Omega} \sum_{\substack{k_1 k_2 q \\ \sigma}} \underbrace{V_{\parallel}(q)}_{\text{same spin}} c^{+}_{k_1+q\sigma} c^{+}_{k_2-q\sigma} c_{k_2\sigma} c_{k_1\sigma}$$

$$+ \underbrace{V_{\perp}(q)}_{\text{opposite spins}} c^{+}_{k_1+q\sigma} c^{+}_{k_2-q-\sigma} c_{k_2-\sigma} c_{k_1\sigma}$$

Coulomb $V_{\parallel} = V_{\perp}$

Contact $V_{\parallel} = 0$

perturbation $H' = \int d\bar{r} \left(\lambda_{\uparrow}(r,t)\, \rho_{\uparrow}(r) + \lambda_{\downarrow}(r,t)\, \rho_{\downarrow}(r) \right)$

$$H_{int} = \frac{1}{2\Omega} \sum_{\substack{k_1 k_2 q \\ \sigma}} \Big(V_{\parallel}(q)\, c^{+}_{k_1+q\sigma} c_{k_1\sigma}\, c^{+}_{k_2-q\sigma} c_{k_2\sigma}$$

$$+ V_{\perp}(q)\, c^{+}_{k_1+q\sigma} c_{k_1\sigma}\, c^{+}_{k_2-q-\sigma} c_{k_2-\sigma} \Big)$$

$$= \frac{1}{2} \sum_{\sigma} \int dr_1 dr_2 \Big\{ V_{\parallel}(r_1-r_2)\, \rho_{\sigma}(r_1)\, \rho_{\sigma}(r_2)$$

$$+ V_{\perp}(r_1-r_2)\, \rho_{\sigma}(r_1)\, \rho_{-\sigma}(r_2) \Big\}$$

let now $\rho_{\sigma}(r) = \underbrace{\bar{\rho}_{\sigma}(r)}_{\text{average induced density}} + \underbrace{n_{\sigma}(r)}_{\text{fluctuation}}$

$$n_{\sigma} = \rho_{\sigma} - \bar{\rho}_{\sigma}$$

e.g. $\int dr_1 dr_2\, V_{ab}(r_1-r_2) \left(\bar{\rho}_a + n_a \right) \left(\bar{\rho}_b + n_b \right) =$

$$= \int dr_1 r_2\, V_{ab}(r_1-r_2) \Big(\bar{\rho}_a \bar{\rho}_b + \bar{\rho}_a \underbrace{n_b}_{\rho_b - \bar{\rho}_b} + \bar{\rho}_b \underbrace{n_a}_{\rho_a - \bar{\rho}_a} + \underbrace{n_a n_b}_{\cancel{0}} \Big)$$

$$\approx \int dr_1 dr_2\, V_{ab}(r_1-r_2) \left(-\bar{\rho}_a \bar{\rho}_b + \bar{\rho}_a \rho_b + \bar{\rho}_b \rho_a \right)$$

Thus mean-field potential energy of a/b-particles

$$H^{MF}_{int,b} = \int dr_2\, \rho_b(r_2) \int dr_1\, V_{ab}(r_1 - r_2)\, \bar{\rho}_a(r_1) = \int dr\, \rho_b(r)\, \gamma_{ab}$$

$$H^{MF}_{int,a} = \int dr_1\, \rho_a(r_1) \int dr_2\, V_{ab}(r_1 - r_2)\, \bar{\rho}_b(r_2) =$$

$$= \int dr\, \rho_a(r)\, \gamma_{ab}(r)$$

$$\gamma_{ab}(r) \equiv \int dr_2\, V_{ab}(r_1 - r_2)\, \bar{\rho}_b(r_2)$$

$$H' = \int dr\, \lambda_a(r,t)\, \rho_a(r)$$

$$\Rightarrow H^{MF}_a = \int dr \left(\gamma_{ab}(r,t) + \lambda_a(r,t)\right) \rho_a(r)$$

$\Rightarrow$ in linear response $\bar{\rho}_a(r,t) = \int dr_1\, dt_1\, \chi_0(r - r_1, t - t_1) \cdot$

$$\cdot \left(\gamma_{ab}(r_1,t_1) + \lambda_a(r_1,t_1\right.$$

$$\bar{\rho}_a(q,\omega) = \chi_0(q,\omega)\left(\lambda_a(q,\omega) + \gamma_{ab}(q,\omega)\right)$$

but $\gamma_{ab}(q,\omega) = V_{ab}(q)\, \bar{\rho}_b(q,\omega)$

$$\bar{\rho}_a(q,\omega) = \chi_0(q,\omega)\left(\lambda_a(q,\omega) + V_{ab}(q)\, \bar{\rho}_b(q,\omega)\right)$$

In case of single species (spinless electrons)

$$\bar{\rho}(q,\omega) = \chi_0(q,\omega)\left(\lambda(q,\omega) + V(q)\, \bar{\rho}(q,\omega)\right)$$

$$\bar{\rho}\,(1 - V\chi_0) = \chi_0 \lambda \Rightarrow \bar{\rho}(q,\omega) = \frac{\chi_0(q,\omega)\, \lambda(q,\omega)}{1 - V(q)\, \chi_0(q,\omega)}$$

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$$\approx \int dr_1 dr_2\, V_{ab}(r_1-r_2) \left(-\bar{\rho}_a \bar{\rho}_b + \bar{\rho}_a \rho_b + \bar{\rho}_b \rho_a \right)$$

Thus mean-field potential energy of a/b-particles

$$H^{MF}_{int,b} = \int dr_2\, \rho_b(r_2) \int dr_1\, V_{ab}(r_1 - r_2)\, \bar{\rho}_a(r_1) = \int dr\, \rho_b(r)\, \gamma_{ab}$$

$$H^{MF}_{int,a} = \int dr_1\, \rho_a(r_1) \int dr_2\, V_{ab}(r_1 - r_2)\, \bar{\rho}_b(r_2) =$$

$$= \int dr\, \rho_a(r)\, \gamma_{ab}(r)$$

$$\gamma_{ab}(r) \equiv \int dr_2\, V_{ab}(r_1 - r_2)\, \bar{\rho}_b(r_2)$$

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$$\Rightarrow H^{MF}_a = \int dr\, \left(\gamma_{ab}(r,t) + \lambda_a(r,t)\right) \rho_a(r)$$

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but $\gamma_{ab}(q,\omega) = V_{ab}(q)\, \bar{\rho}_b(q,\omega)$

$$\bar{\rho}_a(q,\omega) = \chi_0(q,\omega)\left(\lambda_a(q,\omega) + V_{ab}(q)\, \bar{\rho}_b(q,\omega)\right)$$

In case of single species (spinless electrons)

$$\bar{\rho}(q,\omega) = \chi_0(q,\omega)\left(\lambda(q,\omega) + V(q)\, \bar{\rho}(q,\omega)\right)$$

$$\bar{\rho}\,(1 - V\chi_0) = \chi_0 \lambda \Rightarrow \bar{\rho}(q,\omega) = \frac{\chi_0(q,\omega)\, \lambda(q,\omega)}{1 - V(q)\, \chi_0(q,\omega)}$$

Spinfull electrons

$$\gamma_\uparrow(q,\omega) = V_{\parallel}(q)\,\bar{\rho}_\uparrow(q,\omega) + V_\perp(q)\,\bar{\rho}_\downarrow(q,\omega)$$

$$\gamma_\downarrow(q,\omega) = V_{\parallel}(q)\,\bar{\rho}_\downarrow(q,\omega) + V_\perp(q)\,\bar{\rho}_\uparrow(q,\omega)$$

$$\Rightarrow \quad \bar{\rho}_\uparrow = \chi_0\,(\lambda_\uparrow + \gamma_\uparrow)$$

$$\bar{\rho}_\downarrow = \chi_0\,(\lambda_\downarrow + \gamma_\downarrow)$$

where $\chi_0(r,t) = -i\,\theta(t)\langle[\bar{\rho}_\uparrow(r,t), \bar{\rho}_\uparrow(0,0)]\rangle$

response function of the system $\underline{a}$ (or, $\underline{b}$) particles only!

$$\bar{\rho}_\uparrow = \chi_0\,\lambda_\uparrow + \chi_0\left(V_{\parallel}\bar{\rho}_\uparrow + V_\perp\bar{\rho}_\downarrow\right)$$

$$\bar{\rho}_\downarrow = \chi_0\,\lambda_\downarrow + \chi_0\left(V_{\parallel}\bar{\rho}_\downarrow + V_\perp\bar{\rho}_\uparrow\right)$$

$$\begin{pmatrix}\bar{\rho}_\uparrow\\ \bar{\rho}_\downarrow\end{pmatrix} = \chi_0\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\begin{pmatrix}\lambda_\uparrow\\ \lambda_\downarrow\end{pmatrix} + \chi_0\begin{pmatrix}V_{\parallel} & V_\perp\\ V_\perp & V_{\parallel}\end{pmatrix}\begin{pmatrix}\bar{\rho}_\uparrow\\ \bar{\rho}_\downarrow\end{pmatrix}$$

$$\begin{pmatrix}1-\chi_0 V_{\parallel} & -\chi_0 V_\perp\\ -\chi_0 V_\perp & 1-\chi_0 V_{\parallel}\end{pmatrix}\begin{pmatrix}\bar{\rho}_\uparrow\\ \bar{\rho}_\downarrow\end{pmatrix} = \chi_0\begin{pmatrix}\lambda_\uparrow\\ \lambda_\downarrow\end{pmatrix}$$

$$\begin{pmatrix}\bar{\rho}_\uparrow\\ \bar{\rho}_\downarrow\end{pmatrix} = \chi_0\,(\quad)^{-1}\begin{pmatrix}\lambda_\uparrow\\ \lambda_\downarrow\end{pmatrix}$$

$$(\quad)^{-1} = \frac{1}{(1-\chi_0 V_{\parallel})^2 - (\chi_0 V_\perp)^2}\begin{pmatrix}1-\chi_0 V_{\parallel} & \chi_0 V_\perp\\ \chi_0 V_\perp & 1-\chi_0 V_{\parallel}\end{pmatrix}$$

Charge response: $\lambda_\uparrow = \lambda_\downarrow = -\mu$

$$\begin{pmatrix} \bar{\rho}_\uparrow \\ \bar{\rho}_\downarrow \end{pmatrix} = \frac{\chi_0}{(1-V_{\parallel}\chi_0)^2 - (V_\perp \chi_0)^2} \underbrace{\begin{pmatrix} 1-\chi_0 V_{\parallel} & V_\perp \chi_0 \\ V_\perp \chi_0 & 1-V_{\parallel}\chi_0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}}_{\begin{pmatrix} 1-V_{\parallel}\chi_0 + V_\perp \chi_0 \\ 1-V_{\parallel}\chi_0 + V_\perp \chi_0 \end{pmatrix}} (-\mu)$$

$$\begin{pmatrix} \bar{\rho}_\uparrow \\ \bar{\rho}_\downarrow \end{pmatrix} = \frac{\chi_0 \quad (-\mu) \quad (1-V_{\parallel}\chi_0 + V_\perp \chi_0)}{(1-V_{\parallel}\chi_0 - V_\perp \chi_0)(1-V_{\parallel}\chi_0 + V_\perp \chi_0)} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\Rightarrow \bar{\rho}_\uparrow = \bar{\rho}_\downarrow = \frac{\chi_0}{1-(V_{\parallel}+V_\perp)\chi_0} (-\mu) \Rightarrow \chi_e = \frac{\chi_0}{1-(V_{\parallel}+V_\perp)\chi_0}$$

Spin response $\lambda_\downarrow = -\lambda_\uparrow$

$$\Rightarrow \begin{pmatrix} 1-V_{\parallel}\chi_0 & V_\perp \chi_0 \\ V_\perp \chi_0 & 1-V_{\parallel}\chi_0 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \lambda_\uparrow = \begin{pmatrix} 1-(V_{\parallel}+V_\perp)\chi_0 \\ -1+(V_{\parallel}+V_\perp)\chi_0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \left(1-(V_{\parallel}+V_\perp)\chi_0\right)\begin{pmatrix} 1 \\ -1 \end{pmatrix} \lambda_\uparrow$$

$\Rightarrow$ then $$\begin{pmatrix} \bar{\rho}_\uparrow \\ \bar{\rho}_\downarrow \end{pmatrix} = \frac{\chi_0 \lambda_\uparrow}{1-(V_{\parallel}-V_\perp)\chi_0} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$\bar{\rho}_\downarrow = -\bar{\rho}_\uparrow$ $\qquad \chi_s = \dfrac{\chi_0}{1-(V_{\parallel}-V_\perp)\chi_0}$

Imagine contact interaction $\Rightarrow V_{\parallel}(q) = 0$

$$\chi_c = \frac{\chi_0}{1 - V_\perp \chi_0}$$

$$\chi_s = \frac{\chi_0}{1 + V_\perp \chi_0}$$

$$\chi_0(q,\omega) = \frac{1}{\Omega} \sum_k \frac{f(\xi_k) - f(\xi_{k+q})}{\omega + \xi_k - \xi_{k+q} + i\delta}$$

1206.6797
1210.6320

$$\chi_0(q\to 0, \omega = 0) = \frac{1}{\Omega} \sum_k \frac{f(\xi_k) - f(\xi_{k+q} - \xi_k + \xi_k)}{\xi_k - \xi_{k+q}} =$$

$$= \frac{1}{\Omega} \sum_k \frac{\frac{\partial f}{\partial \xi}(\xi_k - \xi_{k+q})}{\xi_k - \xi_{k+q}} = \frac{1}{\Omega} \sum_k \frac{\partial f}{\partial \xi} =$$

$$= \int d\xi\, n(\xi)\left(-\delta(\xi)\right) = -n_0$$

$$f(\xi) = \frac{1}{e^{\beta\xi} + 1} \qquad \frac{\partial f}{\partial \xi} = \frac{-1 \cdot \beta e^{\beta\xi}}{\left(e^{\beta\xi} + 1\right)^2} = -\beta \frac{e^{\beta\xi}}{\left(e^{\beta\xi/2}\left(2\,\mathrm{ch}\frac{\beta\xi}{2}\right.\right.}$$

$$= -\frac{\beta}{4\,\mathrm{ch}^2\left(\frac{\beta\xi}{2}\right)}$$

$$\chi_0(q\to 0, \omega = 0) = \int d\xi\, \frac{1}{2\pi^2} \ln\left(\frac{16}{|\xi|}\right) \frac{-\beta}{4\,\mathrm{ch}^2\left(\frac{\beta\xi}{2}\right)} =$$

$$= -\frac{\beta}{8\pi^2} \frac{2}{\beta} \int_0^\infty dx\, \ln\left(\frac{16\beta}{x}\right) \frac{1}{\mathrm{ch}^2(x/2)} = -\frac{1}{4\pi^2} \ln(16\beta) \cdot 2$$

More accurately, at $T>0$:

$$\frac{\partial f}{\partial \xi} = \frac{\partial}{\partial \xi} \frac{1}{e^{\beta\xi}+1} = -\frac{\beta e^{\beta\xi}}{(e^{\beta\xi}+1)^2} \qquad \xi = \varepsilon - \mu$$

$$\chi_0(q\to 0, 0) = \int d\xi\, N(\xi) \frac{\partial}{\partial \xi}\left(\frac{1}{e^{\beta\xi}+1}\right) \approx$$

$$\cong N(0) \int_{-\mu}^{\mu} d\xi \frac{\partial}{\partial \xi}\left(\frac{1}{e^{\beta\xi}+1}\right) = N_0\left(\frac{1}{e^{\beta\mu}+1} - \frac{1}{e^{-\beta\mu}+1}\right)$$

$$= N_0\left(\frac{e^{-\beta\mu/2}}{2\cosh\left(\frac{\beta\mu}{2}\right)} - \frac{e^{+\beta\mu/2}}{2\cosh\left(\frac{\beta\mu}{2}\right)}\right) =$$

$$= -N_0 \frac{\sinh(\beta\mu/2)}{\cosh(\beta\mu/2)} = -N_0 \tanh\left(\frac{\beta\mu}{2}\right)$$

$\Rightarrow$ instability if $1 + V_\perp\left(-N_0 \tanh\left(\frac{\beta\mu}{2}\right)\right) = 0$

1) $T=0 \Rightarrow \beta = \infty \Rightarrow \tanh\left(\frac{\beta\mu}{2}\right) = 1$

$$1 = V_\perp N_0 \quad \text{(Stoner criterion)}$$

$\Rightarrow$ need finite interaction strength: if $V_\perp > \frac{1}{N_0}$

then instability occurs at finite T_c.

if $V_\perp < 1/N_0 \Rightarrow$ no instability.

(Fermi-liquid state)

2) Suppose $V_\perp > \frac{1}{N_0} \Rightarrow$ can find T_c:

$$\tanh\left(\frac{\beta_c \mu}{2}\right) = \frac{1}{N_0 V_\perp} < 1$$

$$\frac{e^{\beta_c \mu/2} - e^{-\beta_c \mu/2}}{e^{\beta_c \mu/2} + e^{-\beta_c \mu/2}} = \frac{e^{\beta_c \mu} - 1}{e^{\beta_c \mu} + 1} = \frac{1}{N_0 V_\perp}$$

$$e^{\beta_c \mu} - 1 = \frac{1}{N_0 V_\perp}\left(e^{\beta_c \mu} + 1\right)$$

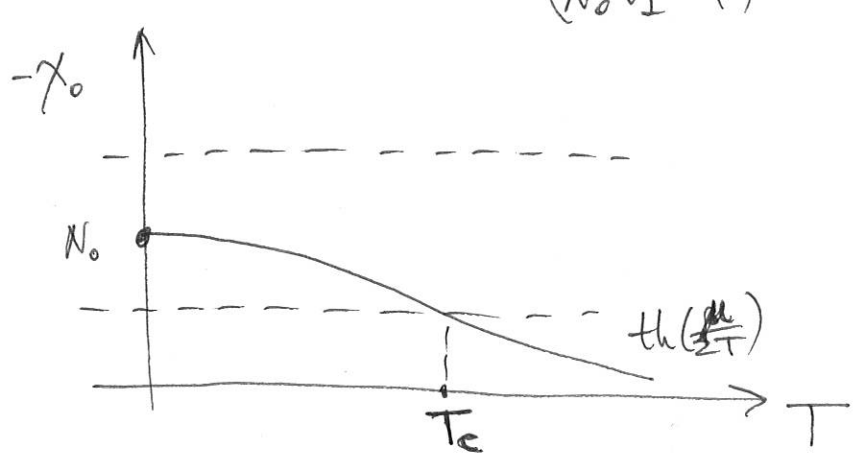
$$\left(1 - \frac{1}{N_0 V_\perp}\right) e^{\beta_c \mu} = 1 + \frac{1}{N_0 V_\perp}$$

$$e^{\beta_c \mu} = \frac{N_0 V_\perp + 1}{N_0 V_\perp - 1} = \cancel{\frac{1 + N_0 V_\perp}{1 - N_0 V_\perp}}$$ possible for $N_0 V_\perp > 1$.

$$\Rightarrow \frac{\mu}{T_c} = \ln\left(\frac{N_0 V_\perp + 1}{N_0 V_\perp - 1}\right)$$

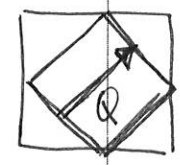
$$T_c = \frac{\mu}{\ln\left(\frac{N_0 V_\perp + 1}{N_0 V_\perp - 1}\right)}$$

This is estimate of crit. temperature



Nesting

$$\chi_0(Q=(\pi,\pi),\omega=0) = \frac{1}{\Omega}\sum_k \frac{f(\xi)-f(-\xi)}{\xi-(-\xi)} \Rightarrow$$



$$\xi_{k+Q} = -\xi_k$$

$$f(-\xi) = \frac{1}{e^{-\beta\xi}+1} = \frac{e^{\beta\xi}}{1+e^{\beta\xi}} = 1-f(\xi)$$

$$f(\xi)-f(-\xi) = \frac{1-e^{\beta\xi}}{1+e^{\beta\xi}} = \frac{-sh(\beta\xi/2)}{ch(\beta\xi/2)}$$

$$\Rightarrow \int d\xi \underbrace{n(\xi)}_{\frac{1}{2\pi^2}\ln\left(\frac{16}{|\xi|}\right)} \left(-th\left(\frac{\beta\xi}{2}\right)\right)\frac{1}{2\xi} =$$

$$= -\frac{2}{2}\int_0^\Lambda d\xi\, n(\xi)\,\frac{th(\beta\xi/2)}{\xi} = -\int_0^{\Lambda\beta/2} dx\, \frac{1}{2\pi^2}\ln\left(\frac{16\beta/2}{x}\right)\frac{th\,x}{x}$$

$$x = \xi\beta/2$$

$$= -\frac{1}{2\pi^2}\ln(8\beta)\int_0^{\Lambda\beta/2} dx\,\frac{th\,x}{x} \sim -\frac{1}{2\pi^2}\ln^2\left(\frac{c}{T}\right)$$

But $V(Q) = -4V \Rightarrow$ for $V>0$

$$1 = 4V\frac{1}{2\pi^2}\ln^2\left(\frac{c}{T}\right)$$

$$\sqrt{\frac{\pi^2}{2V}} = \ln\frac{c}{T} \Rightarrow T = c\,e^{-c'/\sqrt{V}}$$